DO NOW — RECALL

Algorithms: synoptic exam challenge

Name: _____ Class: _____ Date: _____

1. True remaining cost is 4; is h(A)=7 admissible?

2. If g=3 and h=4, what is f?

3. What condition is needed for Dijkstra's standard shortest-path guarantee?

4. When partition sets the pivot at index 2, which recursive index ranges remain within indexes 0–4?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. 7 is an overestimate.

2. Model answer

7.

3. Model answer

Non-negative edge weights.

4. Model answer

0–1 and 3–4.

Vocabulary — know these terms

partition — Rearrange around a pivot and place that pivot.

tentative distance — Current best-known cost from the start.

predecessor — Vertex immediately before another on a path.

heuristic — Estimate of remaining distance to the goal.

admissibility — A heuristic never exceeds the true cheapest remaining cost.

NEW LEARNING

Read, annotate and explain

Match the algorithm to the task

Quicksort partitions an array. Dijkstra finds shortest routes on non-negative weighted graphs. A* adds a remaining-cost estimate.

Show all required intermediate states

For quicksort write arrays after partition. For shortest paths maintain distances, predecessors and a clear visited or frontier order.

Explain efficiency and preconditions

State when each algorithm is appropriate and how its method changes work.

Exam command words matter

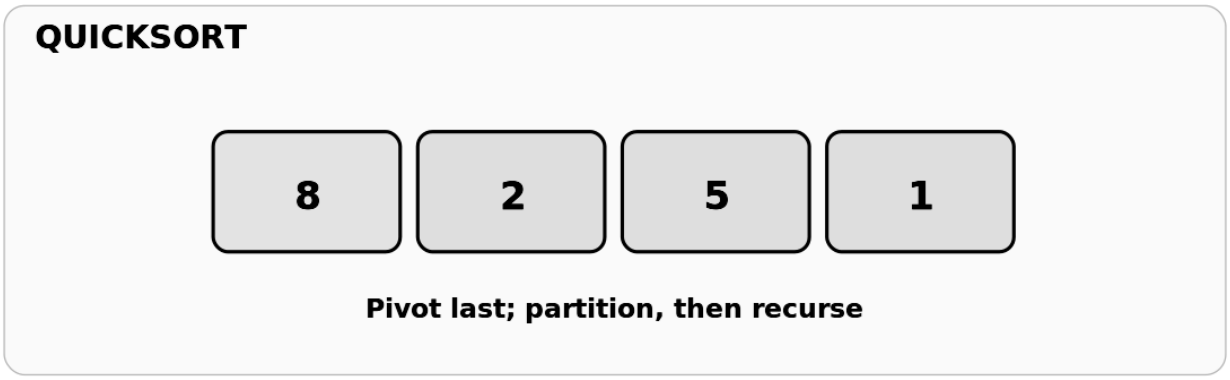
Trace = show states. Explain = give a cause or reason. Compare = connect similarities and differences with evidence.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

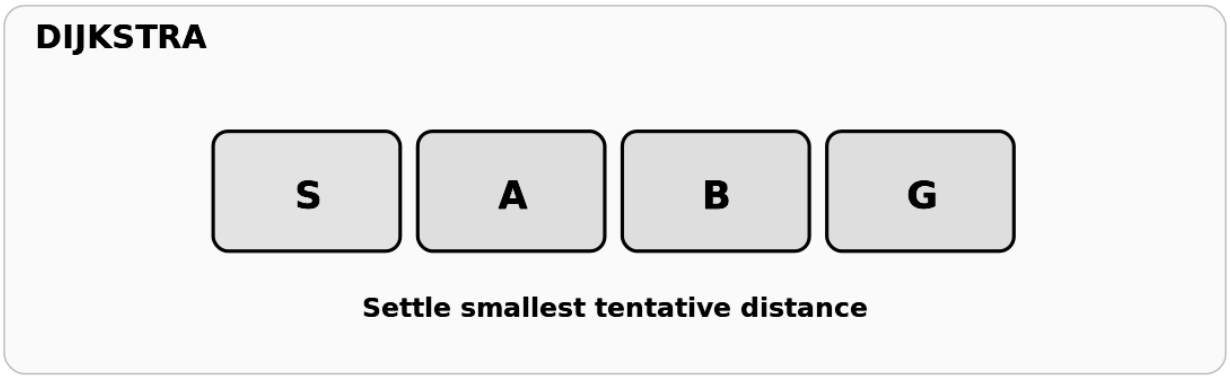
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



QUICKSORT: 8, 2, 5, 1; Pivot last; partition, then recurse

New Learning

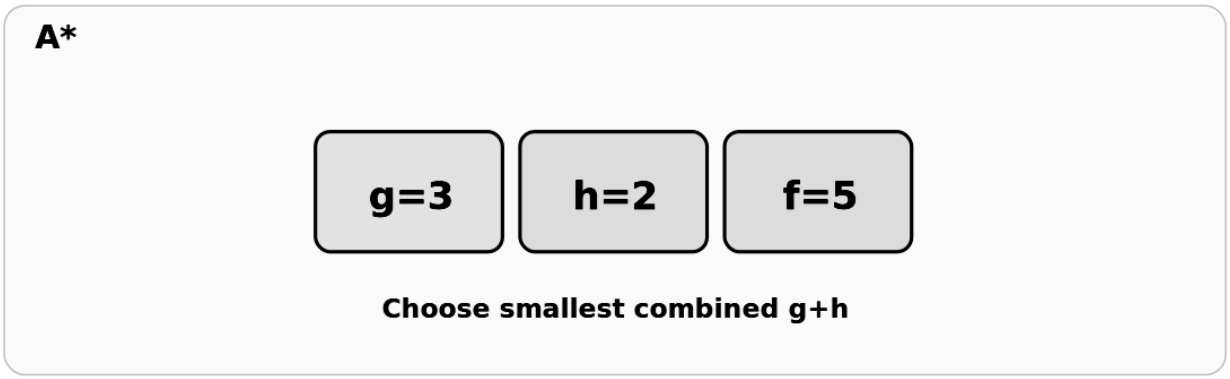


DIJKSTRA: S, A, B, G; Settle smallest tentative distance

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



A*: g=3, h=2, f=5; Choose smallest combined g+h

I DO — WORKED EXAMPLE

Follow the worked method

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

Predict first

Start with pivot 4; 5 is too large to move to the smaller section.

Model response

Move 2 to the left: [2,5,4].

Model response

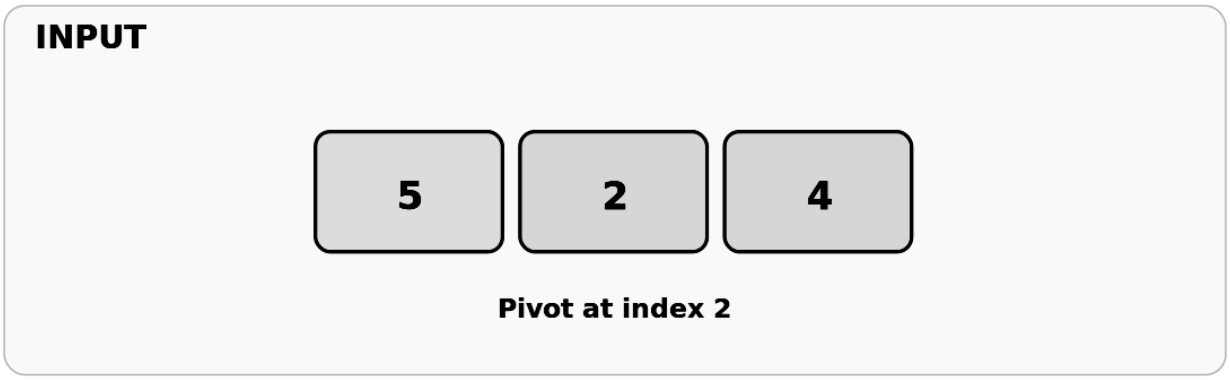
Swap pivot into index 1: [2,4,5]. Only one partition is being demonstrated.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

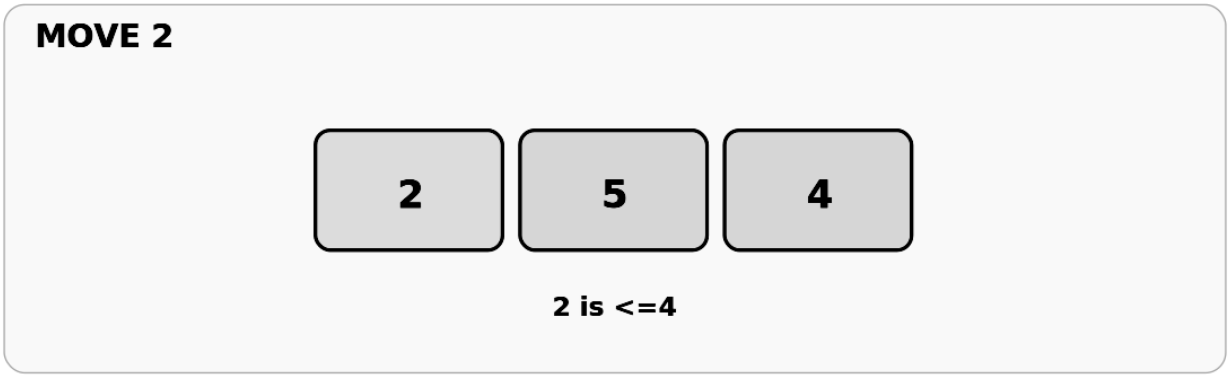
Start with pivot 4; 5 is too large to move to the smaller section.



INPUT: 5, 2, 4; Pivot at index 2

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

Start with pivot 4; 5 is too large to move to the smaller section.



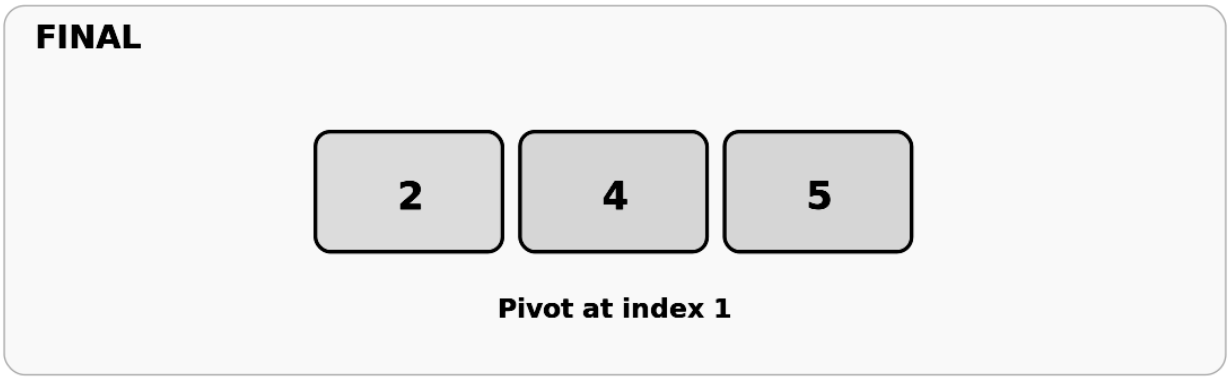
MOVE 2: 2, 5, 4; 2 is <=4

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

Start with pivot 4; 5 is too large to move to the smaller section.



FINAL: 2, 4, 5; Pivot at index 1

WE DO — GUIDED PRACTICE

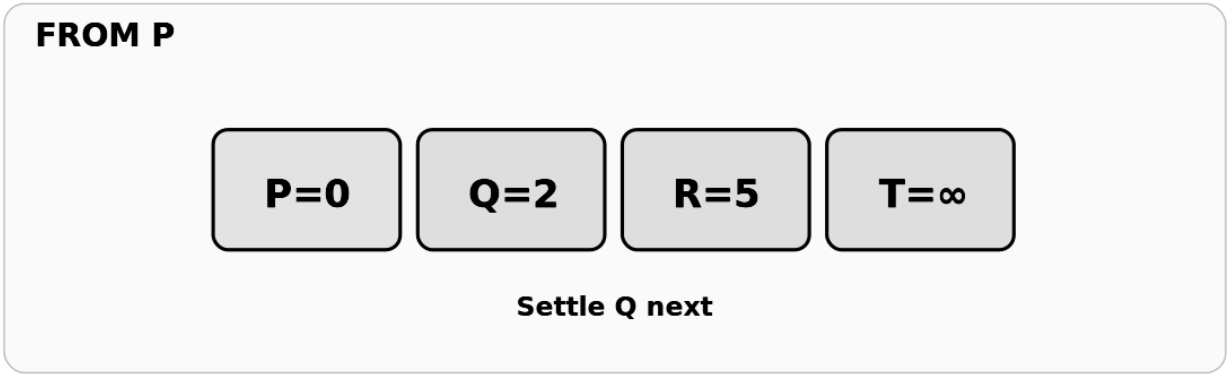
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find P→T shortest route on P–Q=2, P–R=5, Q–R=1, Q–T=8, R–T=3.

From P, Q gets 2 and R gets 5; choose Q.



FROM P: P=0, Q=2, R=5, T=∞; Settle Q next

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first From P, Q gets 2 and R gets 5; choose Q.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Using last-element Lomuto quicksort on [7,2,6,1,5]: (a) state initial pivot [1]; (b) show the array after moving each qualifying value [3]; (c) give array and pivot index after partition [2]; (d) give fully sorted array [2]; (e) state typical average and worst-case time complexity [2].

Question 2

Find the shortest route A→E for non-negative edges: A–B=2, A–C=6, B–C=1, B–D=7, C–D=2, D–E=3, C–E=9. Give tentative-distance improvements [8] and final path [2].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Graph edges S–A=2, S–B=5, A–B=1, B–G=3, A–G=7. Heuristics h(S)=5,h(A)=3,h(B)=2,h(G)=0. (a) Define g, h, f [3]; (b) show initial f for A and B [2]; (c) update B after expanding A [2]; (d) state shortest route and cost [2]; (e) evaluate h(A) admissibility [1].

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Quicksort, Dijkstra and A* are interchangeable because they all have a loop.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

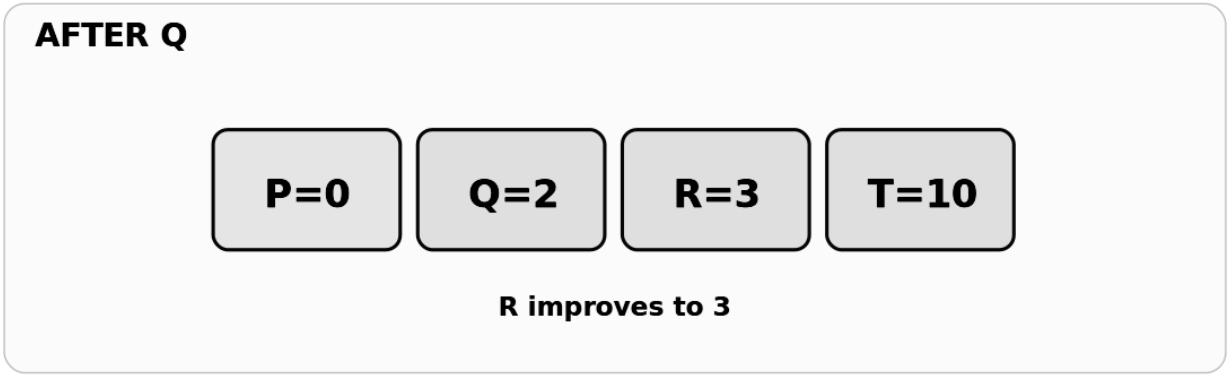
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find P→T shortest route on P–Q=2, P–R=5, Q–R=1, Q–T=8, R–T=3.

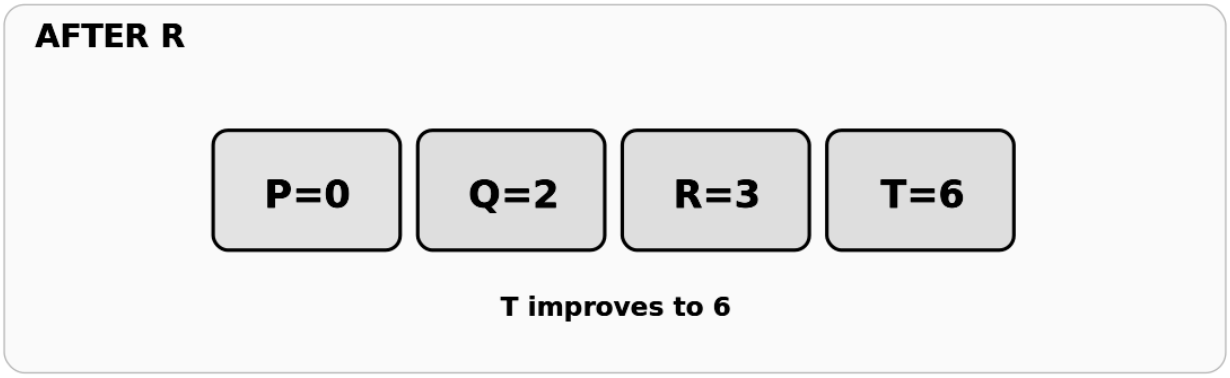
From P, Q gets 2 and R gets 5; choose Q.



AFTER Q: P=0, Q=2, R=3, T=10; R improves to 3

Find P→T shortest route on P–Q=2, P–R=5, Q–R=1, Q–T=8, R–T=3.

From P, Q gets 2 and R gets 5; choose Q.



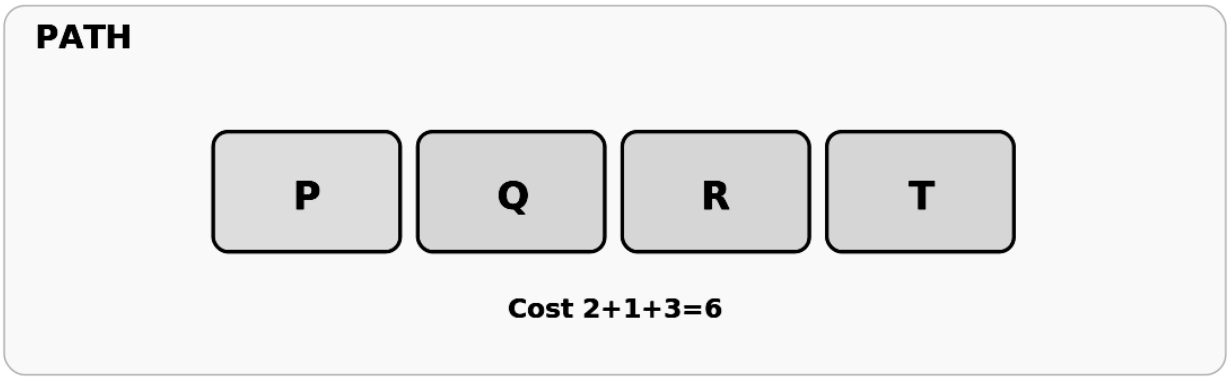
AFTER R: P=0, Q=2, R=3, T=6; T improves to 6

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find P→T shortest route on P–Q=2, P–R=5, Q–R=1, Q–T=8, R–T=3.

From P, Q gets 2 and R gets 5; choose Q.



PATH: P, Q, R, T; Cost 2+1+3=6

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model response

From Q, R improves to 3 and T gets 10.

From R, T improves to 6. The shortest P→T path is P→Q→R→T, total cost 6.

IMPROVE: Explain one correction to your own reasoning.